Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Ultra-Slender Radiation-Hard Probe System for In-Situ Characterization of Spent Nuclear Fuel			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power (✓), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		Mechanical & Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Nuclear Power	Radiation Management Technologies	Nano/Micro Mechanical Systems	System Characterization & Reliability Evaluation Technologies
Concept of Proposed Technology	Nuclear fuel performance, burnup, and isotopic evolution are governed by **probabilistic nuclear reaction processes**, rather than deterministic outcomes at the individual fuel-rod level. As a result, spent nuclear fuel assemblies contain **localized isotopic variations** that cannot be fully quantified by predictive modeling alone. Current fuel reprocessing and waste-management pathways **lack an in-situ quantification step** capable of directly measuring these localized isotopic distributions prior to downstream handling, introducing uncertainty into material characterization and safeguards assumptions. Existing nondestructive assay (NDA) techniques for spent nuclear fuel are mature and widely deployed for safeguards verification and assembly-level characterization; however, they rely on externally deployed detectors and integrated radiation signatures that inherently limit spatial resolution. As documented by the Japan Atomic Energy Agency, current NDA methods			

	provide bulk or assembly-averaged metrics and are not capable of resolving localized or rod-level isotopic variations within intact fuel assemblies [1]. Similarly, Sandia National Laboratories reports that advanced NDA systems are designed around external measurement geometries and device-level outputs, and are not intended to interrogate internal fuel structure or provide localized isotopic mapping [2]. As a result, spent nuclear fuel management and transport decisions remain dependent on probabilistic modeling and integrated external measurements rather than direct in-situ verification. This project develops a **new class of miniature, radiation-hard inspection probes** capable of navigating the 2–3 mm inter-rod gaps inside spent nuclear fuel assemblies within opened dry-storage casks. The probe integrates **compact gamma and neutron detectors**, including diamond, CZT, SPND, and miniature fission-chamber technologies, within a slender, ruggedized housing engineered to survive high-radiation environments. A **robotic delivery system** positions the probe along the length of the fuel assembly, enabling direct measurement of localized isotopic signatures. This approach enables **rod-level isotopic mapping** without disassembling fuel or removing assemblies from their structural configuration.
Rationale for Joint R&D Project	Korean SMEs provide strengths in precision micro-fabrication, compact high-reliability sensing hardware, and robotic mechanisms, all essential to producing a manufacturable 2–3 mm probe system. The U.S. research partner contributes modeling expertise, nuclear measurement capability, and access to a linear accelerator for surrogate radiation-field testing. This division of labor enables rapid prototype development and rigorous validation—work that neither partner could complete alone. The collaboration accelerates a technology with significant future export and safeguards relevance. The recent revision of the U.S.–Republic of Korea 123 Agreement represents a significant evolution in bilateral nuclear cooperation, expanding the scope for joint research, advanced fuel-cycle technologies, and nonproliferation-aligned innovation while maintaining strict safeguards and oversight. Within this framework, there is a timely opportunity to pursue collaborative R&D that strengthens transparency, reduces proliferation risk, and enhances technical understanding of spent nuclear fuel (SNF) without introducing enrichment, reprocessing, or fissile material production pathways. This proposed work aligns squarely with the spirit and letter of the revised agreement by focusing on **measurement, verification, and risk reduction**, rather than fuel fabrication or material separation. By developing advanced in-situ characterization tools and controlled conditioning concepts, the project contributes to shared U.S.–Korean objectives: improving safeguards confidence, enabling better-informed policy and engineering decisions, and advancing export-relevant nuclear technologies that emphasize safety and accountability. Korea's strong industrial base in precision manufacturing, sensors, robotics, and radiation-tolerant electronics complements U.S. strengths in nuclear modeling, accelerator facilities, and safeguards science. This partnership enables development of technologies that are globally relevant but politically and technically challenging for any single nation to pursue alone.
Key Words	1. *Radiation probe* 2. *High-flux sensors* 3. *Spent nuclear fuel characterization* 4. *Robotic inspection* 5. *Nonproliferation*

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	• The project will design and fabricate a **≤3 mm radiation-hard probe** integrating both gamma and neutron detection capabilities. • A **robotic insertion tool** compatible with PWR geometries, and extensible to BWR geometries in later phases, will be developed to enable precise probe positioning within spent fuel assemblies. • The project will establish **radiation-tolerant electronics and data-acquisition systems** required for localized isotopic analysis in high-background environments. • Probe performance will be **validated using controlled surrogate irradiation** at the U.S. partner's linear accelerator facility. Although the integrated probe system is novel, its **core sensing materials, radiation-hard electronics, and robotic insertion technologies** have been demonstrated at laboratory scale in adjacent nuclear, high-energy physics, and industrial inspection applications. The proposed probe architecture leverages **established gamma and neutron detection principles** that have been extensively validated in nuclear measurement applications and adapts them to an ultra-slender, radiation-hardened form factor. [3]			
R&D Target Specifications	• Target specifications include a **probe diameter of 2–3 mm** suitable for insertion into inter-rod gaps within spent fuel assemblies. • The probe must demonstrate **survivability in mixed gamma–neutron radiation fields** potentially as high as $>10^9$–10^{10} n/cm²·s. • The detection system must provide **sufficient spectral sensitivity** to identify key isotopic features under high-background radiation conditions. • The robotic delivery system must achieve **millimeter-scale positioning accuracy** along the full axial height of a fuel assembly. • **Diamond-based detector materials** may be required due to their demonstrated radiation hardness and stability under high gamma and neutron fluence conditions, stable operation at elevated temperatures (150-500 °C) and fast charge collection & low leakage current due to the ultra-wide bandgap (5.5 eV). [4]			
R&D Plan	Consistent with program structure, work will occur in two phases. The first phase will occur during the initial 6-month cycle where teams will agree on project scope. The second phase will be the long-duration cycle where teams design and deliver a functional ultra-slender probe, validated using controlled surrogate irradiation. Phase I will consist of team determination of probe design parameters. If a detailed plan for SNF consolidation is released prior to the start of the project, our team will select design constraints from casks that will be among the earliest planned shipments. In the absence of a detailed plan for consolidation of SNF, the teams will down select the product specifications, in the form of a simulated environment, for probe development. Decisions made will include: • Representative radiation flux characteristics, including neutron & gamma maximums • Representative geometry, including rod axial spacings, options for guide supports, etc. • Concepts for detector collimation, such as probe shielding, adjacent baffles, etc.			

Phase II work will focus on three major areas: 1) detector material selection, 2) packaging and shielding / collimation studies, to include mechanical design of the probe housing and 3) robotic insertion and retrieval mechanism. Design work for each area may proceed in parallel as resources allow, culminating in a final validation test where a mock fuel assembly of interest is used to demonstrate the capability to insert the probe into position, characterize a radiation flux simulating relevant SNF environments, and removal of the probe. For the purpose of minimizing radioactive material production, exposure to a neutron flux may be limited to a detection-only option without exposing robotic parts or other extraneous material unnecessarily. Heating elements may be used during the mock assembly validation step to account for any local heating effects that may potentially impact movement of the probe.

To clarify the motivation for this plan, we highlight (1) the nonproliferation and safeguards benefits and (2) the technical rationale for the selected detector and probe materials.

(1) Nonproliferation and Safeguards Benefits

A defining challenge of the current nuclear fuel cycle is that spent fuel assemblies contain **significant fissile inventories**, particularly plutonium isotopes and residual U-235, yet are treated operationally as homogeneous objects. Conventional nondestructive assay (NDA) methods provide bulk or assembly-averaged estimates but lack the spatial resolution needed to quantify localized fissile distributions or uncertainties arising from burnup history, operating conditions, or fuel management strategies.

This uncertainty drives conservatism in safeguards, storage, and policy decisions—and contributes to proliferation concern, not because of intent, but because of limited observability.

The proposed work addresses this gap directly by:

- Enabling **rod-level or localized isotopic characterization** without dismantling fuel assemblies.
- Providing a technical pathway to **quantify and potentially reduce fissile content** in a controlled, subcritical, non-separative manner.
- Improving confidence in SNF inventories, thereby **reducing perceived and actual proliferation risk**.

Importantly, all concepts explicitly avoid enrichment, chemical reprocessing, plutonium separation, or breeding. Any conditioning activities are designed to contribute to the **decrease**, not increase, fissile inventories, and to do so within engineered facilities subject to safeguards and regulatory oversight.

(2) Technical Rationale for Detector and Probe Materials

A central element of the proposed work is the development of **ultra-slender (2–3 mm diameter), radiation-hard probes** capable of operating within the extreme radiation environments present inside spent fuel assemblies. While no commercial off-the-shelf probes exist that meet these requirements, several mature detector materials provide a strong technical foundation.

Key detector material considerations include:

Diamond Detectors (sCVD or poly-CVD): Diamond offers exceptional radiation hardness, wide bandgap operation, fast charge collection, and high temperature tolerance. These properties make diamond suitable for mixed gamma–neutron environments and prolonged exposure. Diamond detectors can be fabricated with millimeter-scale active areas and are well suited for integration into compact probe geometries where survivability is critical.

Cadmium Zinc Telluride (CZT): CZT provides room-temperature gamma spectroscopy with good energy resolution and compact crystal size. While CZT requires careful packaging and shielding to manage dose rate and neutron damage, it remains one of the few practical options for spectral gamma measurements in constrained geometries without cryogenic systems.

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Self-Powered Neutron Detectors (SPNDs) and Miniature Fission Chambers: These technologies demonstrate that millimeter-scale, radiation-hard neutron detection is feasible in high-flux environments. Although traditionally single-purpose and fixed in location, their underlying principles can be adapted to movable probe architectures when combined with modern materials and electronics.

Supporting Materials and Integration Requirements: To function in SNF environments, detector materials must be integrated with radiation-tolerant cabling (e.g., mineral-insulated conductors or optical fibers), high-temperature encapsulation, and shielding/collimation structures that fit within the 2–3 mm envelope. The innovation in this project lies not in inventing new detector physics, but in **engineering a survivable, deployable, multi-modal probe system** that integrates these materials into a novel form factor.

The proposed research integrates advanced radiation-hard sensing, precision robotics, accelerator-based validation, and safeguards-oriented data analysis into a unified development effort. By tightly coupling modeling, hardware development, and experimental benchmarking between RPI and the Korean SME partner, the project ensures that design decisions are continuously informed by both physical constraints and manufacturability considerations. This integrated approach accelerates progression from laboratory validation to deployable prototype while maintaining alignment with nonproliferation and regulatory expectations.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Rensselaer Polytechnic Institute (RPI) team will lead the **nuclear science, modeling, and experimental validation** aspects of the project. • RPI will develop **radiation transport and detector-response models** to define expected gamma and neutron signatures within spent nuclear fuel geometries and to establish performance requirements for probe materials. • RPI will design **experimental validation protocols** to ensure technical rigor and relevance to real spent fuel environments. • RPI will conduct **surrogate irradiation experiments** using its linear accelerator facility to benchmark detector survivability, signal fidelity, and background discrimination under representative radiation fields. • RPI will develop and validate **data-processing and inversion algorithms** for localized isotopic characterization, unless already owned by Korean SME partner. • RPI will provide **technical guidance on safeguards relevance, subcriticality constraints, and nonproliferation alignment** throughout the project lifecycle.
Expected R&R of Korean SME Partner	• The Korean SME partner will lead the **engineering design, fabrication, and system integration** of the ultra-slender probe platform. • This includes **mechanical design of the 2–3 mm probe housing**, integration of radiation-hard detector materials, development of high-reliability cabling and encapsulation, and implementation of compact electronics suitable for high-dose radiation environments. • The SME will develop the **robotic insertion and positioning system**, including actuation, guidance, and mechanical interfaces compatible with spent fuel assembly geometries. • The SME will conduct **manufacturability and reliability assessments** to support robust system design. • The SME will contribute to **system packaging and integration** in preparation for future certification and deployment. • Based on project outcomes, the SME will support **pathways toward scale-up and commercialization**, including consideration of regulatory and export requirements.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project will deliver the world's first deployable ultra-slender spent nuclear fuel (SNF) inspection probe capable of operating within intact fuel assemblies to provide localized, high-resolution isotopic characterization. This capability enables a level of direct measurement and verification that has not previously been achievable using existing nondestructive assay techniques, which are currently limited to bulk or assembly-averaged metrics. By introducing in-situ, localized quantification prior to downstream handling or transport, the project addresses a critical gap in the nuclear fuel cycle. From a technical and operational perspective, improved characterization of SNF inventories will enhance waste-management decision-making, support risk-informed transport and storage strategies, and strengthen safeguards confidence. These outcomes directly contribute to recognized priorities for long-term nuclear waste disposition, including reducing uncertainty, improving transparency, and enabling measurement-based assurance rather than reliance on predictive modeling alone [5]. Beyond the immediate technical deliverables, the project establishes a new paradigm for how spent nuclear fuel can be understood and managed within an integrated storage, transport, and disposal system. By combining advanced in-situ measurement with safeguards-aligned conditioning concepts, this project establishes a new technical paradigm for how spent nuclear fuel can be understood and managed. The outcomes support: • Enhanced transparency and confidence in SNF inventories. • Reduced proliferation concern through improved observability and potential fissile reduction. • New export-relevant technologies for nuclear inspection, monitoring, and waste-management infrastructure. **Strategic Impact and Long-Term Value** Strategically, the project positions Korean SMEs at the forefront of next-generation nuclear instrumentation while reinforcing the role of U.S.–Korean collaboration as a driver of practical, technically grounded nonproliferation solutions. In the context of the revised U.S.–Republic of Korea 123 Agreement, this work demonstrates how bilateral cooperation can extend beyond policy alignment to deliver concrete technologies that strengthen global confidence in nuclear material management, advance international nonproliferation norms, and support sustainable growth of the civil nuclear enterprise.

6. References

[1] Japan Atomic Energy Agency (JAEA), JAEA-Review-2015-027, "Review of Nondestructive Assay Techniques for Spent Nuclear Fuel." (Sec. 2 pp 4-10, Sec. 3 pp. 11-18)

[2] Sandia National Laboratories, SAND2014-19460, "Spent Fuel NDA Project: Data Authentication Considerations." (Sec. 1 pp. 1-4, Sect 2 pp. 50-20, Sect 4 pp. 35-50)

[3] Knoll, G. F., Radiation Detection and Measurement, 4th Edition, Wiley. (Ch 2, 10, and 15)

[4] Shimaoka, T., Koizumi, S., J. H., & Kaneko. (2021). Recent progress in diamond radiation detectors. *Functional Diamond*, *1*(1), 205–220. https://doi.org/10.1080/26941112.2021.2017758 [pp. 205-207, 211-214]

[5] U.S. Department of Energy, Strategy for the Management and Disposal of Used Nuclear Fuel and High-Level Radioactive Waste, January 2013. (DOE, 2013, pp. 3, 5-7, 9, 12)

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization